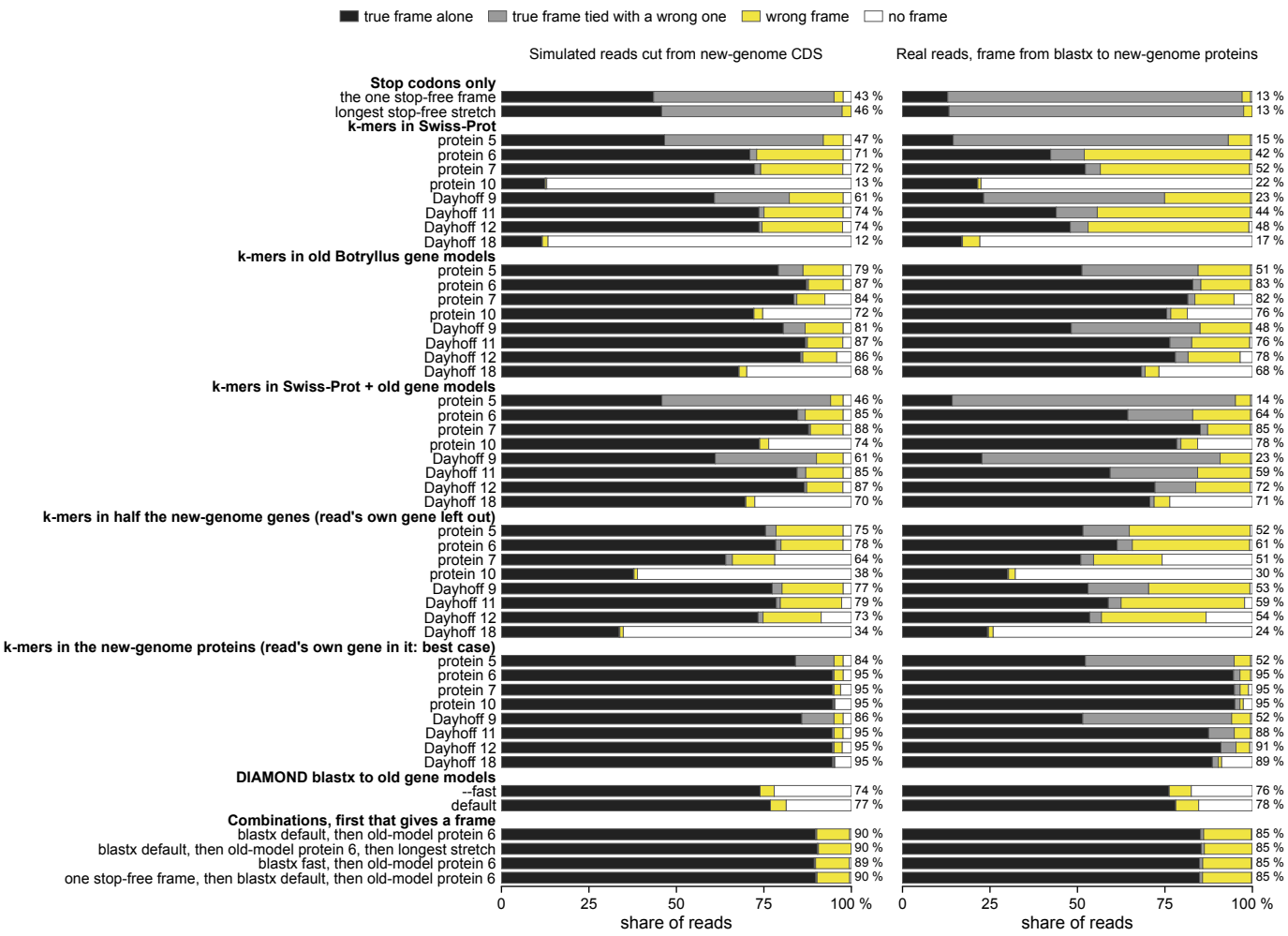
true frame alone
true frame tied with a wrong one
wrong frame
no frame
Simulated reads cut from new-genome CDS
Real reads, frame from blastx to new-genome proteins
Stop codons only
the one stop-free frame
longest stop-free stretch
k-mers in Swiss-Prot
protein 5
protein 6
protein 7
protein 10
Dayhoff 9
Dayhoff 11
Dayhoff 12
Dayhoff 18
k-mers in old Botryllus gene models
protein 5
protein 6
protein 7
protein 10
Dayhoff 9
Dayhoff 11
Dayhoff 12
Dayhoff 18
k-mers in Swiss-Prot + old gene models
protein 5
protein 6
protein 7
protein 10
Dayhoff 9
Dayhoff 11
Dayhoff 12
Dayhoff 18
k-mers in half the new-genome genes (read's own gene left out)
protein 5
protein 6
protein 7
protein 10
Dayhoff 9
Dayhoff 11
Dayhoff 12
Dayhoff 18
k-mers in the new-genome proteins (read's own gene in it: best case)
protein 5
protein 6
protein 7
protein 10
Dayhoff 9
Dayhoff 11
Dayhoff 12
Dayhoff 18
DIAMOND blastx to old gene models
--fast
default
Combinations, first that gives a frame
blastx default, then old-model protein 6
blastx default, then old-model protein 6, then longest stretch
blastx fast, then old-model protein 6
one stop-free frame, then blastx default, then old-model protein 6
43 %
46 %
47 %
71 %
72 %
13 %
61 %
74 %
74 %
12 %
79 %
87 %
84 %
72 %
81 %
87 %
86 %
68 %
46 %
85 %
88 %
74 %
61 %
85 %
87 %
70 %
75 %
78 %
64 %
38 %
77 %
79 %
73 %
34 %
84 %
95 %
95 %
95 %
86 %
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95 %
74 %
77 %
90 %
90 %
89 %
90 %
13 %
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15 %
42 %
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44 %
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17 %
51 %
83 %
82 %
76 %
48 %
76 %
78 %
68 %
14 %
64 %
85 %
78 %
23 %
59 %
72 %
71 %
52 %
61 %
51 %
30 %
53 %
59 %
54 %
24 %
52 %
95 %
95 %
95 %
52 %
88 %
91 %
89 %
76 %
78 %
85 %
85 %
85 %
85 %
0
25
50
75
100 %
share of reads
0
25
50
75
100 %
share of reads